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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Using the image already available in Colab
img = cv2.imread('image1.jpg', 0)

if img is not None:
    # Apply Sobel Y edge detection
    sobel_y = cv2.Sobel(img, cv2.CV_8U, 0, 1, ksize=5)

    # Save and display the result
    cv2.imwrite('sobel_y.jpg', sobel_y)
    print("Sobel Y Result:")
    cv2_imshow(sobel_y)
else:
    print("Error: 'image1.jpg' not found. Please upload the image file to (
```

Sobel Y Result:

